PHY202 Practice Exam on chapter(s): 18, 19

1. $\dot{N}_e = 2.02\times10^{20}$ electrons per second $\because \Delta V = \frac{\Delta PE}{q}, W = \Delta PE$ for a conservative force, so $W = -q\Delta V$ and so $\frac{q}{t} = \frac{-W}{t\Delta V}$ and so finally $\dot{N}_e = \frac{q}{q_e t} = \frac{-W}{q_e t\Delta V}$

2. $q = 3.76\times10^{-8}\,\mathrm{C} \because \Sigma\vec{F}_x = m\vec{a}_x, F = k\frac{|q_1 q_2|}{r^2}$

3. $F = 7.38\times10^{-6}\,\mathrm{N} \because F = Eq$

4. $\theta = 89.9$□° west of south $\because E = k\frac{|Q|}{r^2}$

5. $v_f = 4.30\times10^{6}\frac{\mathrm{m}}{\mathrm{s}} \because \Delta V = \frac{\Delta PE}{q}, W = -\Delta PE$ for a conservative force, so $W = -q\Delta V, W_{net} = \Delta KE$ so $\Delta KE = -q\Delta V$

6. $C_T = 2.73\times10^{-9}\,\mathrm{F} \because \frac{1}{C_s} = \frac{1}{C_1} + \ldots + \frac{1}{C_n}$

7. $W_{ratio} = 107.$ times more energy $\because \Delta V = \frac{\Delta PE}{q}, W = -\Delta PE$ for a conservative force, so $W = -q\Delta V$

8. $\Delta V = 7.93\times10^{3}\,\mathrm{V} \because V = \frac{kQ}{r}$

9. $C_T = 2.31\times10^{-9}\,\mathrm{F} \because \frac{1}{C_s} = \frac{1}{C_1} + \ldots + \frac{1}{C_n}, C_p = C_1 + \ldots + C_n$

10. $\Delta V_{max} = 6.30\times10^{9}\,\mathrm{V} \because \Delta V = Ed$

11. $F = 9.78\times10^{-6}\,\mathrm{N} \because F = k\frac{|q_1 q_2|}{r^2}$

12. $x_3 = 0.658\,\mathrm{m} \because \Sigma\vec{F}_x = m\vec{a}_x, F = k\frac{|q_1 q_2|}{r^2}$

13. $F = 14.4\,\mathrm{N} \because F = Eq, \Delta V = Ed \because$ to $1e^-$ the gun supplies $E = 43.\,\mathrm{keV}$, its $\Delta V = 43.\,\mathrm{kV}$.

14. $F = 2.73\times10^{-13}\,\mathrm{N} \because F = Eq$

15. $E = 1.40\times10^{6}\,\mathrm{N/C} \because E = k\frac{|Q|}{r^2}$

16. $Q = 2.20\times10^{-10}\,\mathrm{C} \because C = \frac{Q}{V} = \epsilon_0\frac{A}{d}$

17. $C = 2.16 \times 10^{-5}\,\mathrm{F}$ $\because E_{cap} = \dfrac{CV^2}{2}$

18. $Q = 7.05 \times 10^{-6}\,\mathrm{C}$ $\because V = \dfrac{kQ}{r}$